

SELECTION, NOT PERCEPTION: RE-RANKING DETECTOR CANDIDATES FOR SHEET-IMAGE SCORE FOLLOWING ON REAL RECORDINGS

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ABSTRACT

Real-time score following on sheet-music images needs no symbolic score, but systems trained on synthesized audio lose accuracy on real recordings. We show that for detection-based trackers this loss is a failure of selection rather than perception: on real room-microphone piano recordings a correct candidate is among the detector’s top 256 boxes at every scored onset, and a causal oracle over them tracks 99.6% of onsets within 0.5 s where the detector’s argmax tracks 80.0%. We therefore freeze the detector and learn only to choose. CANDOR combines a parameter-free transition prior with a 23k-parameter set-based candidate scorer trained on synthesized audio alone. Applied without retraining to three released detectors it raises accuracy on the real recordings by 20.3, 11.0 and 7.9 points, reaching 91.6% on the weakest and 94.4% on the strongest, above a recent end-to-end system that additionally reads layout annotations at test time, at 1.25 ms per step on one CPU thread. Every constant is chosen by leave-one-piece-out; a variant with no tuned constants at all reaches 88.7%. We further report that synthetic validation accuracy is nearly uninformative about real-audio accuracy ($r = 0.16$ over 48 checkpoints), that three different image representations give identical real-audio results, and that the prior which buys these gains also costs accuracy after score discontinuities.

Index Terms— Score following, sheet music, real-time alignment, candidate re-ranking, music information retrieval

1. INTRODUCTION

Score following, the real-time alignment of a live performance to its score, underlies automatic page turning [1] and computer accompaniment [2, 3]. Most music exists only as printed or scanned sheets, and optical music recognition remains error-prone [4], so a line of work follows the performance directly on the sheet image, mapping audio and a score page to the current position [5, 6, 7, 8, 9, 10, 11].

These systems are trained on audio synthesized from the score, and real recordings remain their main open problem. On the public real piano performances aligned to sheet images [7], the detection-based CYOLO-SB [10] drops from 89.3% of onsets tracked within 0.5 s on synthetic audio to 79.9%, and CUNet from 85.5% to 22.4% [11]. Proposed remedies change

the training: impulse-response augmentation [9], extra proprietary recordings [10], or a new cascaded architecture trained end to end that also reads system and bar layout annotations at test time [11].

We start from a measurement instead. The CYOLO-SB detector proposes 8,112 candidate note boxes per step and the tracker reports the most confident one. On the real room-microphone recordings, a box within the 0.5 s tolerance is among the 256 most confident candidates at every one of the 4,149 scored onsets, and an oracle that knows the answer but must commit one step at a time and may never move backwards tracks 99.6% of them. The argmax tracks 80.0% (Fig. 1). The information needed to follow the performance survives the room; what fails is the rule that reads a position out of it. For these models, robustness to real audio is a selection problem.

Our contributions are: (i) a diagnosis, by candidate-set oracles, that locates the synthetic-to-real gap of sheet-image trackers in selection rather than perception; (ii) CANDOR, a decoder for a *frozen* detector combining a parameter-free transition prior with a 23k-parameter set-based scorer, which lifts three separately released detectors without retraining any of them, needs no layout annotations, and runs an order of magnitude faster than the cascade it outperforms; (iii) three measurements that constrain how such systems should be built and evaluated: synthetic validation barely ranks models for real audio, the image representation does not matter, and the transition prior that produces the gains is also what makes discontinuities hard.

Related work. Symbolic score followers infer the position with hidden (semi-)Markov models [12, 13] or online time warping [14]. Sheet-image trackers moved from staff strips [5] and reinforcement-learning agents [6, 7] to full-page segmentation [8] and audio-conditioned object detection [9, 10] with FiLM [15]. CODA [11] adds a system, bar and note cascade with learned transition penalties and beam search, trained end to end with layout supervision, and contributes a jump benchmark we adopt in Sec. 3.4. We keep the detector and re-rank its output, akin to learned re-scoring of detections [16, 17]. Because the scorer conditions on its own past choices, it faces the exposure bias of sequence prediction [18, 19].

2. METHOD

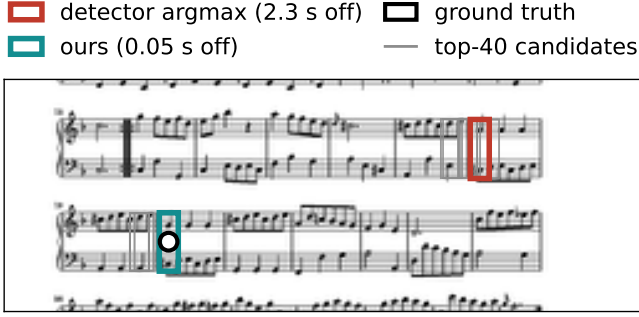


Fig. 1. One onset of a real room-microphone recording. Grey: the detector’s 40 most confident note candidates. The argmax (red) sits on the wrong staff; CANDOR (teal) picks a lower-ranked box on the true note (ring). The correct box was proposed but not chosen, the typical failure on real audio.

2.1. Candidates from a frozen detector

A performance is processed in steps $t = 1, \dots, T$. At each step the detector receives the audio up to t and the current page and outputs note boxes $b_i = (x_i, y_i, w_i, h_i)$ with objectness o_i on three anchors of a 52×52 grid. We keep the $K = 256$ most confident, C_t . Staves are concatenated along one horizontal axis, the unrolled coordinate u_i , so that positions on either side of a line break are close. The baseline outputs

$$\hat{u}_t = \arg \max_{i \in C_t} o_i. \quad (1)$$

Everything below reads only C_t and the decoder’s own previous choice $\hat{u}_{t-1} = u_{\hat{u}_{t-1}}$. No detector weight is updated and no layout annotation is read.

2.2. Transition prior

Between steps Δ_t frames apart, the performer moves forward by a roughly predictable amount. We score the displacement $d_i = u_i - \hat{u}_{t-1}$ with a floored Gaussian whose mean grows with elapsed time,

$$\log \pi_i = \max \left(-\frac{(d_i - \mu \rho_t)^2}{2\sigma^2}, J \right), \quad (2)$$

where $\rho_t = \text{clip}(\Delta_t/\Delta_0, 0.2, 8)$ and $\Delta_0 = 5$ frames is the mean inter-onset gap in training. The floor J keeps jumps possible at a bounded cost. The three constants (μ, σ, J) are not learned; Sec. 3.1 states how they are chosen. Alone, the hand score $h_i = \log o_i + \log \pi_i$ already improves every detector we tried (Table 2).

2.3. Candidate scorer

Each candidate is described by $\phi_i \in \mathbb{R}^{33}$: confidence (log objectness, gap to the frame’s best, rank); box geometry; motion relative to the history (normalized displacement, backward and

staff-change flags, residual to a constant-velocity extrapolation); relation to the detector’s predicted system and bar boxes when it emits them; and a neighbourhood group describing whether the box looks like a real notehead. We also read the 128-dimensional feature f_i from which the detector’s head scores the box. A permutation-invariant network [20] embeds the set and pools a context shared by all candidates:

$$e_i = g([\phi_i; \text{ReLU}(W f_i)]), \quad c = \left[\frac{1}{K} \sum_j e_j; \max_j e_j \right], \quad (3)$$

$$s_i = r([e_i; c]), \quad (4)$$

where g and r are two-layer perceptrons (64, 32 and 64, 1 units), $W \in \mathbb{R}^{64 \times 128}$, and the maximum is per dimension, so each candidate is judged against the others in its frame. The scorer has 23,203 parameters, 1.6% of the 1.46M-parameter detector.

2.4. Training and decoding

We run the frozen detector over the MSMD training pieces [21], synthesized and convolved with measured room impulse responses [22], and store C_t and f_i at every onset. With ε_i the error in frames of candidate i , the soft target is $q_i \propto \exp(-\varepsilon_i/\tau)$ with $\tau = 3$, and the loss is the listwise cross-entropy

$$\mathcal{L} = - \sum_{i \in C_t} q_i \log \frac{\exp s_i}{\sum_{j \in C_t} \exp s_j}. \quad (5)$$

History features come from the best candidates of the previous steps (teacher forcing), perturbed on 30% of samples by Laplace noise of scale 30 px. We train with AdamW (learning rate $3 \cdot 10^{-3}$, cosine schedule, 40 epochs, batch 256) and keep the epoch with the best greedy rollout on the validation pieces. At test time the learned and hand scores are blended,

$$\hat{u}_t = \arg \max_{i \in C_t} \beta s_i + (1 - \beta) h_i, \quad (6)$$

and $\hat{u}_t$ becomes the history of step $t + 1$. Decoding is causal and greedy: one forward pass over 256 candidates per step, 1.25 ms on a single CPU thread, against 12.8 ms on a GPU for the cascade of [11].

A variant with no constants at all. Training the scorer to predict the residual $s_i - h_i$ and decoding $\arg \max_i (s_i + h_i)$ is Eq. 6 at $\beta = 0.5$ and removes β from the system. Fixing (μ, σ, J) in the training objective removes the other three. That variant tunes nothing on any evaluation set and is reported in Table 2.

3. EXPERIMENTS

3.1. Setup

Data. The detectors are the three released checkpoints [9, 10]. Scorer are trained on each detector’s own candidates

over the 353 MSMD training pieces and validated on the 19 validation pieces. The test set is the real-performance subset of [7] (Setting II in [11]): 16 pieces, 25 pages and 4,149 onsets played on a Yamaha AvantGrand N2 hybrid piano and recorded with a room microphone. A held-out set uses 80 pieces withheld from the scorer’s training, rendered with a real room impulse response and additive noise at 12 to 0.5 dB SNR (40,594 onsets per level).

Metric and protocol. As in [10, 11] we report the ratio of onsets whose predicted position maps to within 0.05 to 5 s of the true time, with 0.5 s as the primary figure, and 95% intervals from a bootstrap over pieces [23].

How the four constants are chosen. (μ, σ, J, β) are selected by leave-one-piece-out: for each of the 16 test pieces they are fitted on the other 15 with pieces weighted equally, and that piece is then decoded with them. No piece contributes to the constants used to decode it. Every number we report for CANDOR is cross-validated this way, and the no-constant variant of Sec. 2 needs no selection at all. Retraining with another seed moves the score by about a point, so all rows are seed means with standard deviations.

3.2. CANDOR transfers across detectors

Table 1 places CANDOR beside the published systems on the same recordings, and Table 2 decomposes it. Our run of each released detector reproduces its published argmax to within 0.001.

The same decoder, retrained only on each detector’s own candidates, improves all three. It is worth 20.3 points on CYOLO, the weakest and the one whose training data is entirely public, lifting it past the strongest released detector’s starting point. On CYOLO-SB+A it reaches 94.4%. Both open-data rows already exceed CODA’s 88.3%, which retrains the whole model and reads layout annotations at test time, and CANDOR does so at a tenth of the per-step cost with no detector training at all.

Two rows guard the result. The zero-parameter prior alone accounts for roughly half the gain on CYOLO-SB and CYOLO-SB+A, so the learned component is not carrying the claim by itself. And the variant with no tuned constants whatsoever reaches 88.7%, still above CODA, which is the number to read if the leave-one-piece-out protocol is not accepted.

3.3. What does not matter, and what does

The image representation does not matter. The lower block of Table 2 feeds the scorer the detector’s own backbone features, DINOv2 patch features [26] at the same grid, and nothing at all. All three land on 90.9. On the synthetic validation split the same three differ by 3.4 points (95.1 featureless against 98.5), so the features are learnable and useful there and worth nothing on real audio.

History does matter. Removing the features computed against the tracker’s previous position costs 7.2 points, the

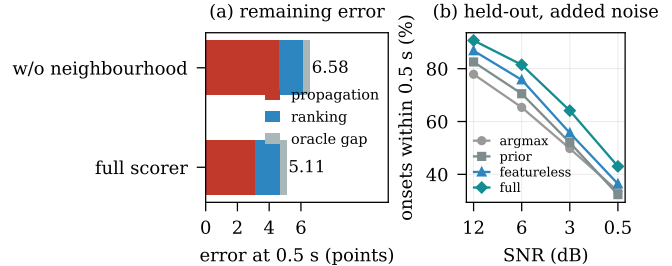


Fig. 2. (a) Error on the real recordings at 0.5 s, split by teacher forcing into error propagation, ranking, and the gap to a causal oracle over the same candidates. (b) Held-out pieces with a real room impulse response and additive noise (40,594 onsets per level).

largest single ablation in the table, and it costs 4.2 points on validation as well. Teacher forcing attributes about two thirds of CANDOR’s remaining error to propagation: supplied the true previous position the same scorer reaches 98.0% and the causal oracle 99.6%.

The audio encoder does not rescue perception either. Replacing the detector’s mel-CNN audio tower with MERT [24] and retraining costs more than 40 points, and **Mamba, the encoder of [11], result pending**. Neither approaches what re-ranking the unchanged candidates achieves.

Synthetic validation barely ranks models. Across 48 trained checkpoints, accuracy on the synthetic validation split correlates $r = 0.16$ with accuracy on the real recordings. The single best-validation checkpoint (98.98) scores 89.3 on real audio, below the featureless scorer. A held-out reverberant set does better ($r = 0.80$) but its top pick is also mediocre. This is why every constant here is cross-validated on the target recordings rather than chosen on a synthetic proxy, and it is a caution for the field: model selection on synthesized audio is close to arbitrary for this task.

3.4. Score discontinuities

CANDOR’s prior assumes the performance moves forward, which is what makes it work and also what breaks it at a repeat or a jump. We adopt the random subset of the benchmark of [11]: three jumps spliced into each performance at random destinations, the harder of their two subsets. We splice the audio and re-run the detector, so the post-jump conditioning is genuinely wrong for two seconds; we exclude the jumps that cross a page boundary, where our decoder resets its state and would recover for free, leaving 41.

Table 3 is a negative result reported in full. CANDOR is *worse* than the detector’s argmax after a jump, 0.15 against 0.22 of onsets tracked in the following five seconds, because the prior pins the tracker to a stale position. Porting the silence-triggered break of [11] recovers most of that when a break is present and nothing when it is absent, exactly as its design

Table 1. Real piano recordings (16 pieces, 4,149 onsets, room microphone). Ratio of onsets tracked within each error threshold. Published rows are from Table 1 of [11], which also lists weaker systems [5, 7, 8]; [†] detector trained with additional proprietary recordings, public checkpoint. *Layout*: system and bar annotations needed at test time. Our rows freeze the detector of the row above them and are cross-validated per piece.

Method	Onset error threshold (s)					Layout	Retrains detector
	≤0.05	≤0.1	≤0.5	≤1.0	≤5.0		
CYOLO [9]	.563	.581	.712	.749	.919	no	—
CYOLO-SB [10]	.610	.630	.799	.832	.960	no	—
CYOLO-SB+A [†] [10]	.682	.706	.865	.891	.981	no	—
CODA [11]	.721	.743	.883	.914	.987	yes	yes
CYOLO + CANDOR (ours)	.	.	.916	.	.	no	no
CYOLO-SB + CANDOR (ours)	.	.	.909	.	.	no	no
CYOLO-SB+A [†] + CANDOR (ours)	.	.	.944	.	.	no	no

Table 2. Each row is the same decoder on a different frozen detector, or the same detector with a different scorer, all cross-validated per piece on the real recordings at 0.5 s. n is the number of training seeds.

Configuration	n	argmax	+prior	+scorer
<i>Frozen detector</i>				
CYOLO [9]	3	71.3	72.5	91.6 ± 1.4
CYOLO-SB [10]	6	80.0	86.5	90.9 ± 0.6
CYOLO-SB+A [†]	3	86.5	90.8	94.4 ± 0.9
<i>Decoder variants, on CYOLO-SB</i>				
no tuned constants at all	3	80.0	86.5	88.7 ± 0.9
no history features	3	80.0	86.5	83.7 ± 0.4
<i>Audio encoder replaced, detector retrained</i>				
MERT [24]	3	33.4	31.9	46.2 ± 9.1
Mamba [25], as [11]	3	.	.	.
<i>Image features for the scorer, on CYOLO-SB</i>				
detector backbone (default)	6	80.0	86.5	90.9 ± 0.6
DINOv2 [26]	1	80.0	86.5	90.9
none	1	80.0	86.5	90.9

predicts. A trigger on the tracker’s own confidence instead of on silence works in both cases but only returns to argmax. None approaches CODA’s 0.71. Discontinuity recovery is where an end-to-end cascade earns what it costs, and a frozen-detector re-ranker does not get it for free.

3.5. Robustness and cost

On the held-out pieces (Fig. 2b) CANDOR improves on the prior alone at every noise level, and the margin shrinks as the SNR falls: the scorer helps most where the detector’s boxes are still reliable. On the 94 synthetic test pieces of Setting I, where the published detector scores .893 at 0.5 s, CANDOR reaches .955, matching CODA. Reducing the candidate budget from $K = 256$ to 128 costs 0.5 points; the correct box sits at

Table 3. Jump recovery, 41 within-page jumps on the real recordings. *rec@k*: fraction of jumps first tracked correctly within k s. *post*: onsets tracked within 1 s over the following 5 s.

Method	silence before jump		no silence	
	rec@1	post	rec@1	post
argmax	0.32	0.22	0.27	0.22
CANDOR	0.12	0.15	0.10	0.15
+ break [11]	0.27	0.19	0.10	0.15
+ confidence trigger	0.15	0.22	0.20	0.23

rank 27 at the 90th percentile, so the tail is doing real work.

4. CONCLUSION

Sheet-image trackers lose accuracy on real recordings mostly because of how they select a position, not what they perceive. A 23k-parameter re-ranker over a frozen detector’s own candidates recovers most of that gap on three separately released detectors, needs no layout annotations and no detector training, and runs ten times faster than the cascade it outperforms. The negative results are as informative: the image representation is irrelevant, the audio encoder does not rescue the gap either, synthetic validation barely ranks models, and the forward-motion prior that makes the method work is exactly what makes discontinuities hard. As in prior work the correct page is given, all performances are piano, and with 16 test pieces differences of a point or two are not resolvable. Agreement between the pitches a candidate implies and those heard is a natural next source of history-independent evidence. Code and trained scorers will be released.

5. COMPLIANCE WITH ETHICAL STANDARDS

This is a numerical study that uses only publicly available data; no ethical approval was required.

6. ACKNOWLEDGMENTS

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